VORLDX INDUSTRIES PRIVATE LIMITED

EMPLOYEE ATTENDANCE & GEOFENCING SYSTEM

Comprehensive System Architecture, Technical Implementation & Verification Report

Document Version: 1.0.0 Production
Date: October 2026
Target Environment: Full-Stack MERN (Node.js, Express, React/Vite, MongoDB)
Status: Fully Implemented & Tested

1. EXECUTIVE SUMMARY

The VorldX Employee Attendance System is an enterprise-grade, secure, full-stack attendance management solution built for modern workplace scenarios (On-site, Hybrid, and Remote). It combines photographic face verification, real-time GPS 50-meter geofencing, single daily attendance rules, automated shift duration tracking, Brevo transactional email OTP verification, and dynamic HR admin governance.

2. SYSTEM ARCHITECTURE & TECH STACK

- **Backend:** Node.js 20+, Express.js framework, CommonJS module architecture, RESTful API conventions.
- **Database:** MongoDB with Mongoose ODM, strictly typed schemas, atomic update hooks, compound unique indexes.
- **Frontend:** React 18, Vite bundler, Tailwind CSS 3 with custom glassmorphism design system, Lucide icons.
- **Security & Auth:** JSON Web Tokens (JWT) in HTTP-only cookies, bcrypt salt hashing, role-based access control (HR Admin vs Employee).
- **Email Engine:** Brevo API / SMTP via nodemailer, verified sender delivery, dynamic 6-digit cryptographic OTP generation.
- **Geolocation & Geofencing:** HTML5 Geolocation API on client, server-side Haversine mathematical distance validation in meters.

3. DEEP IMPLEMENTATION DETAILS

A. 50-Meter GPS Geofencing Rule

To prevent proxy or remote check-in fraud, the system includes a strict 50-meter boundary enforcement around the physical

office coordinates:

- **Coordinate Input:** Admin can input raw latitude/longitude, click "Fetch Current Device GPS", or paste an office Google Maps link.
- **Link Parser:** Regex auto-parses embedded @lat,long, query params, and coordinates directly into system coordinates.
- **Haversine Formula:** Calculates accurate distance in meters between employee device and office GPS coordinates: R = 6371km.
- **Strict Enforcement:** If distance > 50 meters, server rejects attendance with HTTP 403 error specifying exact calculated distance.

B. Single Daily Attendance & Checkout Workflow

- **Single Daily Check-In:** An employee can only check in once per calendar day. Duplicate check-in attempts are prevented atomically.
- **Checkout System:** Once checked in, the employee card displays their check-in time and provides a dedicated "Check Out" action.
- **Shift Duration:** On checkout, total working minutes are calculated and formatted as X hours Y minutes.
- **Day Finalized:** Once checkout is completed, the day is marked as Completed with both timestamps preserved and locked.

C. Face Photo Verification & Scanner

- **Camera Stream:** Webcam / mobile front camera video stream captured with HTML5 canvas with selfie mirroring.
- **Photo Storage:** Captured photo is attached to attendance record and auto-registered as profile photo upon first capture.
- **Admin Verification:** HR Admins can view captured photos in attendance tables and matrix reports.

4. DATABASE DESIGN & DATA MODELS

- **User:** employeeId (unique), name, email, role (hr_admin/employee), workType (office/remote/hybrid), profilePhoto, isActive.
- **AttendanceRecord:** employee (Ref User), date (UTC Start of Day), checkIn, checkOut, status (present/late/wfh), workingMinutes, method, capturedPhoto.
- **AttendancePolicy:** companyTimezone (Asia/Kolkata), officeLocation: { latitude, longitude, address, locationLink, radiusMeters: 50, isEnabled }, shifts.
- **WFHRequest:** employee, requestedDate, reason, status (pending/approved/rejected), reviewer, reviewComment.
- **AuditLog:** actor, action (ATTENDANCE_CHECK_IN, ATTENDANCE_CHECK_OUT, POLICY_UPDATED), targetId, metadata, ipAddress.

5. VERIFICATION & TEST RESULTS

Test Scenario	Expected Behavior	Result
Location Within 50m (22m distance)	Attendance allowed and marked	PASS [OK]
Location Outside 50m (222m distance)	HTTP 403 Forbidden with exact distance error	PASS [OK]
Attendance Without GPS Location	HTTP 403 Location permission required error	PASS [OK]
Single Check-in Per Day Rule	Second check-in attempt blocked	PASS [OK]
Check-Out Execution	Check-out time recorded, working duration calculated	PASS [OK]
Vite Frontend Production Build	Clean compilation, 0 bundle or syntax errors	PASS [OK]

6. PRODUCTION STATUS & SIGN-OFF

All fake demo records (EMP001-EMP005) have been completely cleaned from the production database. The application is running in a clean state with real employee accounts, live OTP delivery, real-time geofence enforcement, and accurate IST date calculation. The system is verified as 100% functional and ready for operational deployment.

System Sign-off:

Certified By: VorldX Automated Quality Assurance & Engineering Team